

a**Client updates** $u_1, \dots, u_n$ **of round** r **Stage 1: geometric-median acceptance region** m = geometric median of all updates (Weiszfeld) $B_0 = \{i: \|u_i - m\| \leq 2.5 \times \text{median distance}\}$ reset $B_0 = U$ if $|B_0| < n/2$ in (b): **Stage 2: relative-density gate** $rd_i = k' / \sum$ of k' -NN distances, $k' = 5$ opens if the largest rd gap $> 10 \times$ the median gapand $\geq \max(2, \lceil 0.05n \rceil)$ Stage-1 rejections; 3-round memory

gate open

Stage 3: cluster validationSNN clustering of the low-density set $L = \{i: rd_i < t\}$ reject group S if $\|c_S - m_{B_0}\| > \tau R_0$, $\tau = 2$ R_0 = median distance to m_{B_0} within B_0 in (b): **Combine:** $B = B_0 \setminus (\text{rejected groups}) \subseteq B_0$ valve: if $|B| < n/2$, restore $B = B_0$ aggregate: $w_{r+1} = w_r + \text{mean of } u_i \text{ over } i \in B$ **b**

Illustrative 2-D example (synthetic points)

sparse honest group

 B_0

Stage-1 boundary

consensus radius τR_0

attackers

Stage 3 rejects 2 groups;
none has a member in B_0 , so $B = B_0$ ● honest, in B ★ geometric median m ○ honest, outside B_0

□ group rejected by Stage 3

▲ attacker, outside B_0 ✕ its centroid c_S